The denominator *problem*

Scope 3 Category 15 is the only emissions figure that moves when the market moves. How financed emissions is calculated, the five places the calculation breaks, and what actually closes the data gap.

IN THIS BRIEFING

9 exhibits · 20 sources · data as at 1 August 2026

FOR

Banks · insurers · asset managers

FRAMEWORKS

PCAF Part A and B · GHG Protocol Category 15 · IFRS S2 · AASB S2

AUGUST 2026 · RGI-2026-03

CONTENTS

What is in this briefing.

ASSUMED KNOWLEDGE

The reader is assumed to know what Scope 3 is. Category 15 is not defined here, and reporting thresholds are not covered — they are documented by the regulators and are not the constraint.

ON SINGLE-SOURCED FIGURES

Named in the sentence rather than only in the endnote. Several widely circulated statistics on portfolio data gaps trace to vendors with a commercial interest in the gap and are excluded rather than repeated.

A figure that moves when the market moves.

> Financed emissions is the only major emissions number where the reporting institution has almost no direct access to the underlying data — and where the reported total can fall while every borrower emits exactly as much as before.

The mechanism is arithmetic, not accounting judgement. Attribution divides an institution's exposure by the counterparty's total value. For listed equity and corporate bonds that denominator is **EVIC**, a market value. When valuations rise the denominator rises, the attribution factor falls, and reported financed emissions fall.[1,8]

Everything downstream inherits the property: targets set against a baseline that moves with markets, year-on-year movements that may be vintage artefacts, and — on the parts of the book with no counterparty data — a figure that is a restatement of sector exposure multiplied by fixed coefficients.

WHAT THIS BRIEFING ARGUES

Financed emissions cannot be read as a single number. The estimate, the method and the data quality have to travel together — and the metric worth managing is not the tonnage but the share of the portfolio for which the tonnage is actually measured.

Six figures.

> Each is a property of the standard as written, not a criticism of any institution applying it.

3

DATA VINTAGES CAN SIT INSIDE ONE RATIO — EXPOSURE, EVIC AND COUNTERPARTY EMISSIONS ARE MEASURED AT DIFFERENT DATES[1]

3–5

THE ONLY DATA QUALITY SCORES AN EMISSION FACTOR DATABASE CAN SUPPLY; SCORES 1 AND 2 REQUIRE COUNTERPARTY DATA[4]

33%

WEIGHTING APPLIED TO FACILITATED EMISSIONS, LEAVING THE REMAINING TWO-THIRDS ATTRIBUTED TO NO ONE[3]

6

ASSET CLASSES CARRY GHG PROTOCOL CONFORMANCE REVIEW; THOSE ADDED SINCE 2020 DO NOT[2]

10

ASSET CLASSES IN THE DECEMBER 2025 THIRD EDITION, FROM SIX IN 2020[1]

2030

REASONABLE ASSURANCE OVER AUSTRALIAN CLIMATE DISCLOSURES, FINANCED EMISSIONS INCLUDED, FROM 1 JULY[15]

The estimate is only as good as the score attached to it.

Financed emissions attributes a share of a counterparty's emissions to whoever funded it: exposure divided by counterparty value, multiplied by that counterparty's emissions.[1] The design is deliberate — equity and debt are weighted equally so that all capital providers to one company sum to its total emissions, and that common denominator is the standard's principal control against double counting.

The difficulty is that the denominator is not one thing. For listed equity it is EVIC, a market value including cash taken at fiscal year end.[8] For business loans it is book equity plus debt. For mortgages and vehicles it is value at origination. For sovereign debt it is PPP-adjusted GDP.[1] Only one of those moves with markets, and it governs the largest listed exposures.

Where counterparty data is absent the calculation falls to economic-activity estimation: revenue multiplied by a sector emission factor, or at the bottom of the scale an exposure converted to implied revenue through sector asset-turnover ratios. At that point the figure has stopped measuring the counterparty and started measuring the sector.[1]

THE CONSEQUENCE FOR MANAGEMENT

On a book scored 4 or 5, financed emissions cannot detect a borrower decarbonising. It moves only when exposure moves — so the reported figure can be reduced by reallocating between sector codes with no change in the real economy. That is why the weighted data quality score and the coverage percentage are the numbers carrying the information.[1]

01

The calculation

One formula, seven denominators and a five-point quality scale. The mechanics matter because every failure in the next section is a property of them.

Exposure over value, times emissions.

Exhibit 1

The whole method is one ratio, and the fragile half sits underneath the line.

Attribution of counterparty emissions to a financial institution, per counterparty.

$$\underset{\text{ATTRIBUTION FACTOR}}{\frac{\text{Outstanding amount}}{\text{Total value of the counterparty}}} \times \underset{\text{SCOPE 1 AND 2}}{\boxed{\text{Counterparty emissions}}} = \underset{\text{SUMMED ACROSS THE PORTFOLIO}}{\boxed{\text{Financed emissions}}}$$

HALF OF THE RATIO	WHO HOLDS IT	MEASURED
Numerator	The institution's own book — a carrying amount it controls and can reconcile	at the reporting date
Denominator	The counterparty, or the market; seven different definitions across the asset classes	varies by asset class

Note: PCAF applies the same attribution principle across asset classes, weighting equity and debt equally, so that all capital providers to one counterparty sum to its total emissions. That is the standard’s principal control against double counting within an institution.

Source: PCAF Part A, Third Edition, December 2025

REGENESIS IMPACT

The numerator is a carrying amount from the institution's own books. The denominator comes from the counterparty and, for the largest asset class, from the market. The two are not measured on the same basis, at the same frequency, or by the same party.

Seven denominators, one formula.

Exhibit 2

Only one denominator moves with markets, and it governs the largest listed exposures.

Denominator by asset class under PCAF Part A, with the basis of measurement.

ASSET CLASS	DENOMINATOR	BASIS
Listed equity and corporate bonds	**EVIC** — market capitalisation of ordinary and preferred shares, plus book value of total debt and minority interests, without deducting cash	market value
Business loans and unlisted equity	Total company equity plus debt	book value
Project finance	Total project equity plus debt	book value
Commercial real estate	Property value at loan origination	at origination
Mortgages	Property value at loan origination	at origination
Motor vehicle loans	Vehicle value at loan origination	at origination
Sovereign debt	PPP-adjusted GDP in international dollars	national accounts

Note: EVIC was proposed by the EU Technical Expert Group on Sustainable Finance and codified in the Benchmark Regulation delegated acts in July 2020; the "including cash" construction avoids treating as cash what the market does not price as cash. PCAF moved the sovereign denominator to PPP-adjusted GDP in 2022, on the reasoning that debt stock is not a proxy for the value of a country.

Source: PCAF Part A; EU Technical Expert Group on Sustainable Finance; World Bank

REGENESIS IMPACT

Five scores, and how far each sits from the borrower.

Exhibit 3

The scale measures distance from the counterparty, not precision of arithmetic.

PCAF data quality score for corporate asset classes, from reported emissions to sector estimation.

1
Reported emissions, verified by a third party
The counterparty measured it and someone independent checked it.

2
Reported emissions unverified, or primary energy-consumption data
Self-calculated by the counterparty, or built from its actual energy use with matched factors.

3
Primary production data
Physical output — tonnes, megawatt hours, square metres — with production-specific factors.

4
Company revenue with a sector emission factor
The counterparty's own revenue, multiplied by tCO_2e per unit of sector revenue.

5
Outstanding amount only
Revenue inferred from the exposure using sector asset-turnover ratios, then the sector factor applied. Two estimation layers.

Note: each asset class has its own table; real estate scores are built on metered building energy, energy performance certificates and floor area rather than company financials. Institutions must disclose a weighted-average score by asset class, weighted by outstanding amount, alongside the percentage of the portfolio covered.

Source: PCAF Part A, Third Edition

REGENESIS IMPACT

What carries conformance, and what does not.

> "PCAF is GHG Protocol conformant" is true of the 2020 standard and its six asset classes. It is not true of everything published since.

Reviewed

The six asset classes in the 2020 first edition — listed equity and corporate bonds, business loans and unlisted equity, project finance, commercial real estate, mortgages, and motor vehicle loans — were reviewed and found in conformance with the GHG Protocol Scope 3 Standard for Category 15.[2,6]

Not reviewed

Sovereign debt and emission removals, added in 2023, and everything added in the December 2025 third edition — sub-sovereign debt, securitisations, use-of-proceeds structures and undrawn commitments. The GHG Protocol has since closed the review service that granted the mark.[1]

WHY IT MATTERS NOW

The GHG Protocol has a live revision of Scope 3, including a discussion paper on Category 15 dated November 2024.[7] An institution building a financed-emissions capability today is building against a standard whose newest components carry no conformance review and whose parent standard is under revision. That is an argument for documenting method choices, not for waiting.

02

Where it breaks

Five failures. Each has a mechanism that can be stated precisely, and each changes what the reported number can be used for.

Reported emissions fall when markets rise.

Exhibit 4

A rising counterparty valuation cuts attributed emissions with no change in the real economy.

Effect of a change in counterparty market value, exposure and physical emissions held constant.

	YEAR 1	YEAR 2, VALUATION UP 40%
Exposure held	100	100 — unchanged
Counterparty EVIC	1,000	1,400
Attribution factor	10.0%	7.1%
Counterparty emissions	1,000 tCO_2e	1,000 tCO_2e — unchanged
Financed emissions reported	100 tCO_2e	**71 tCO_2e**

Note: illustrative arithmetic, not observed data. The mechanism is the standard as written — EVIC is a market value, so the attribution factor is partly a function of price. The effect reverses in a drawdown: reported financed emissions rise in a falling market.

Source: PCAF Part A; EU Technical Expert Group on Sustainable Finance; ReGenesis Impact analysis

REGENESIS IMPACT

PCAF has acknowledged that institutions may apply corrections, and multi-year EVIC averaging is used in practice. Averaging suppresses the variance. It does not change the fact that the denominator is a price.[1]

Three vintages in one ratio.

Exhibit 5

The three inputs to a single year's figure are not measured at the same date.

Typical measurement dates for the components of a financed-emissions disclosure.

COMPONENT	WHERE IT COMES FROM	TYPICAL VINTAGE
Outstanding amount	The institution's own book	reporting date
EVIC	Market data, conventionally at the counterparty's prior fiscal year end	prior year
Counterparty emissions	The counterparty closes its year, compiles an inventory, obtains assurance and publishes; a provider then ingests and maps it	up to two years earlier

Note: the lag is structural rather than incidental — it is the time cost of the counterparty's own reporting cycle plus provider processing. PCAF cites reduced data lag as a benefit of integrating CEDA into its database.

Source: PCAF Part A; PCAF database documentation

REGENESIS IMPACT

WHAT IT DOES TO A TREND

A year-on-year movement can be produced entirely by vintages shifting — a newer emissions dataset arriving, or an EVIC reference date rolling forward — with no change in exposure or in counterparty behaviour. A trend line is not evidence of decarbonisation unless the vintages are held constant or the movement is decomposed.

At the bottom of the scale, the borrower disappears.

Exhibit 6

Economic-activity estimation cannot distinguish two counterparties in the same sector.

What each data option can and cannot detect.

OPTION	DETECTS A COUNTERPARTY DECARBONISING	MOVES WHEN
1 — reported	**Yes**, once the counterparty reports it	counterparty emissions change
2 — physical activity	**Yes**, directly and without reporting lag	energy use or production changes
3a — revenue × sector factor	**No** — intra-sector dispersion is not captured	revenue changes, or the factor is updated
3b — exposure only	**No** — revenue is itself inferred from exposure	exposure or sector classification changes

Note: revenue-based factors are denominated in currency, so inflation, exchange-rate movement and margin change shift the estimate with no physical change. PCAF's 2025 edition recommends applying an inflation adjustment to economic emission factors, and a year-on-year fluctuation analysis to separate real change from mechanical change.

Source: PCAF Part A, Third Edition

REGENESIS IMPACT

Two companies in one sector with a tenfold difference in carbon intensity receive the same estimate per unit of revenue. On a book scored 4 or 5 the reported figure is, in substance, the institution's **sector exposure mix multiplied by fixed coefficients**.

Double counting, three kinds.

Exhibit 7

One of the three is controlled by the standard; the other two are not.

Forms of double counting in financed emissions, and their treatment.

KIND	MECHANISM	TREATMENT
Within an institution	Holding both debt and equity in one counterparty, or lending along a single value chain	controlled by the common denominator
Across institutions	Attribution is designed so all capital providers to one counterparty sum to 100% of its emissions — correct per reporter, but the figures are not additive across the system	inherent
Inside input-output estimates	Multi-regional input-output models used for counterparty Scope 3 overlap statistically; Rabobank finds the overlap depends on the institution's market share, and that the multipliers give no insight into its size	unquantified

Note: PCAF states that certain metrics should not be aggregated for portfolio-level comparison, citing data availability, aggregation and potential double counting. Rabobank proposes that institutions using an input-output model report the statistical double-counting percentage alongside the figure.

Source: PCAF Part A; Rabobank, “Double checking double counting”

REGENESIS IMPACT

Avoided emissions and emission removals must be reported separately and may not be netted against Scope 1, 2 or 3 financed emissions, at facility or portfolio level.[1]

Two-thirds attributed to no one.

> Capital markets activity sits under a separate standard, with a single scalar in place of a measurement — and the gap between the two standards creates an arbitrage neither closes.

100%

ATTRIBUTION BASIS WHEN THE EXPOSURE SITS ON THE BALANCE SHEET, UNDER PART A[1]

33%

WEIGHTING FACTOR APPLIED TO FACILITATED EMISSIONS UNDER PART B, WITH DISCLOSURE OF THE FACTOR REQUIRED[3]

What the factor is

A single judgement applied uniformly, reported as derived from the methodology used to classify globally systemically important banks — not from an estimate of a facilitator's causal contribution to an issuance.[3]

What follows from it

Facilitated emissions are reported separately and are not added to financed emissions. Moving an exposure from the balance sheet to a facilitated structure reduces the attributed share from the whole to a third, with no change in the underlying activity.

03

Closing the gap

What moves a portfolio up the quality scale — and the one route that is closed, which is where most programmes start.

No database reaches above score 3.

Exhibit 8

Emission factor databases supply scores 3 to 5 by construction; 1 and 2 exist only where the counterparty supplied data.

What each source delivers, against the data quality score it can support.

SOURCE	WHAT IT SUPPLIES	SCORE
Counterparty, assured	Reported Scope 1 and 2, third-party verified	1
Counterparty, self-reported or primary energy data	Self-calculated emissions, or metered energy consumption	2
Counterparty, production data	Physical output volumes with matched factors	3
PCAF emission factor database CEDA integrated, around 60,000 factors	Sector and regional factors, including “rest of world” coverage	3–5
Input-output models EXIOBASE, USEEIO	Emissions per unit of spend or revenue, by sector and region	4–5

Note: a separate PCAF European building emission factor database serves real estate portfolios, covering EU countries plus Norway, Switzerland and the United Kingdom. CDP is the principal route to reported data at scores 1 and 2; PCAF and CDP published a joint mapping of their data quality concepts in June 2023.

Source: PCAF database documentation; PCAF and CDP, June 2023; EXIOBASE; US EPA

REGENESIS IMPACT

This is the finding most programmes reach late. Buying more factors improves coverage and consistency. It cannot raise the score above 3, because the score measures distance from the counterparty — and a database is, by construction, not the counterparty.

What actually moves the score.

Exhibit 9

Every route above score 3 runs through the counterparty, which makes collection a programme rather than a request.

Approaches with published precedent, and what each delivers.

APPROACH	WHAT IT DELIVERS	PRECEDENT
Standardised industry questionnaire	A common baseline across lenders, so clients answer once rather than once per bank; MAS points banks to the ABS Environmental Risk Questionnaire as a baseline template for collecting customer information	Singapore[17,16]
Counterparty transition scoring	Converts partial counterparty data into a decision-useful signal; Deutsche Bank operates a 1–7 Transition Maturity Score with defined minimum plan requirements, combining an automated score with manual adjustment	published framework[18]
Reported-data pipelines	Routes counterparty disclosure into the calculation at scores 1 and 2; PCAF and CDP have mapped their data quality concepts onto each other	PCAF and CDP[5]
Proxy governance	Documented decisions on proxy sources, assumptions, methodologies and limitations, substantiated and feeding the next iteration	MAS expectation[16]

Note: MAS sets expectations on the governance of proxy data rather than mandating a measurement method, and frames portfolio financed emissions as a metric banks may choose to use. The obligation to disclose financed emissions sits in the ISSB-aligned reporting regimes rather than in the transition planning guidelines.

Source: MAS transition planning guidelines; Association of Banks in Singapore; Deutsche Bank; PCAF and CDP

REGENESIS IMPACT

The deadline that sets the horizon.

> Australian financial institutions must reach reasonable assurance over financed emissions built on proxy data. That, not the first disclosure date, determines how far data quality has to move and by when.

Yr 1

LIMITED ASSURANCE OVER GOVERNANCE, PARTS OF STRATEGY, AND SCOPE 1 AND 2 ONLY[15]

Yr 2–3

LIMITED ASSURANCE PHASES ACROSS REMAINING DISCLOSURES — WHERE SCOPE 3 AND FINANCED EMISSIONS FIRST ATTRACT IT[15]

2030

REASONABLE ASSURANCE OVER ALL CLIMATE DISCLOSURES, PERIODS COMMENCING ON OR AFTER 1 JULY[15]

A CORRECTION WORTH CARRYING

Financed emissions is **not** an industry-based metric. Under IFRS S2 it sits in the main body at paragraph 29(a)(vi)(2), with guidance at B58–B63.[10] AASB S2 omitted the industry-based metrics requirement but retained the financed-emissions package deliberately,[13] and AASB S2025-1 amends it — which a requirement that did not exist could not be.[12] The weaker "consider the applicability of" wording appeared in Exposure Draft SR1 and was not carried into the final standard.[14]

Five actions.

- **Publish the weighted data quality score and the coverage percentage beside the tonnage.** PCAF requires both by asset class; a total without them cannot be interpreted, and disclosing them is the cheapest credibility available.[1]
- **Decompose any year-on-year movement before presenting it.** Separate exposure change, valuation change, vintage change and real emissions change — the fluctuation analysis PCAF's 2025 edition recommends.[1]
- **Target the score, not the tonnage, on the book scored 4 or 5.** On those exposures the tonnage cannot detect decarbonisation; only moving up the data hierarchy changes what the number is able to show.
- **Treat counterparty data collection as a programme with a standard instrument.** Every route above score 3 runs through the counterparty, and a shared questionnaire reduces client fatigue while producing comparable answers.[17]
- **Document the method choices now, while they are still defensible.** Denominator conventions, EVIC reference dates, averaging, sector mapping and proxy sources all become assurance evidence before 2030.[15,16]

THE ASYMMETRY

Coverage and data quality improve only at the speed counterparties respond, which is measured in reporting cycles. Assurance arrives on a fixed date. The distance between those two clocks is the whole of the problem.

Scope, method and limitations.

Scope. The measurement of Scope 3 Category 15 financed emissions under the PCAF standard, its treatment in IFRS S2 and AASB S2, and the assurance sequence applying in Australia. Reporting thresholds and phase-in dates are out of scope, as are insurance-associated emissions under PCAF Part C.

Method. Compiled from the standards and supporting documentation listed in the endnotes. Where a figure or characterisation rests on a single published source, that source is named in the sentence. Illustrative arithmetic is labelled as such. **Data as at 1 August 2026.**

LIMITATIONS

Three matter. Several widely circulated statistics on the share of portfolios lacking counterparty data trace to vendors with a commercial interest in the size of that gap; they are excluded rather than repeated. The sector-by-sector phase-in of counterparty Scope 3 reporting could not be confirmed and is described only in principle. And this briefing describes the standard as published — it does not report what any institution has disclosed, and the GHG Protocol's revision of Category 15 remains in progress.[7]

Sources.

1 PCAF, The Global GHG Accounting and Reporting Standard for the Financial Industry, Part A, Third Edition
carbonaccountingfinancials.com, December 2025

2 PCAF, Part A, First Edition — the six asset classes reviewed as conformant with the GHG Protocol
carbonaccountingfinancials.com, November 2020

3 PCAF, Part B — facilitated emissions for capital markets
carbonaccountingfinancials.com, December 2023

4 PCAF emission factor database, following integration of CEDA
carbonaccountingfinancials.com, 2025

5 PCAF and CDP, "The importance of data quality in the journey toward decarbonization"
carbonaccountingfinancials.com, June 2023

6 GHG Protocol, Corporate Value Chain (Scope 3) Standard, Technical Guidance chapter 15 — Investments
ghgprotocol.org, 2013

7 GHG Protocol, Scope 3 revision discussion paper C.1 — Investments (Category 15)
ghgprotocol.org, 7 November 2024

8 EU Technical Expert Group on Sustainable Finance; EVIC codified in the Benchmark Regulation delegated acts
ec.europa.eu, July 2020

9 Rabobank, "Double checking double counting: quantifying the overlap in input-output-table-based scope 3 emissions"
rabobank.com, 2024

10 IFRS S2 "Climate-related Disclosures", paragraph 29(a)(vi)(2) and paragraphs B58-B63
ifrs.org, June 2023

11 ISSB, Amendments to IFRS S2 — Amendments to Greenhouse Gas Emissions Disclosures (ISSB/2025/1)
ifrs.org, 11 December 2025

12 AASB S2 "Climate-related Disclosures"; AASB S2025-1 amending the financed-emissions requirements
standards.aasb.gov.au, September 2024; 17 December 2025

13 AASB S2 Knowledge Hub, on industry-based metrics and financed emissions
aasb.gov.au, 2025

14 AASB Exposure Draft SR1, draft paragraphs AusB59.1, AusB61.1 and AusB63.1 — superseded
aasb.gov.au, October 2023

15 AUASB, ASSA 5000 and ASSA 5010 sustainability assurance standards
auasb.gov.au, January 2025

16 MAS, Guidelines on Environmental Risk Management: transition planning
mas.gov.sg, 5 March 2026

17 Association of Banks in Singapore, Environmental Risk Questionnaire
abs.org.sg, 2022

18 Deutsche Bank, Transition Finance Framework
db.com, November 2025

19 EXIOBASE multi-regional environmentally extended input-output database; US EPA USEEIO
exiobase.eu, current

20 World Bank, GDP at purchasing power parity, current international dollars
worldbank.org, current

Figures are quoted as published by the issuing body. Claims that could not be corroborated across independent sources have been excluded rather than qualified.

Who produced this briefing.

ReGenesis Impact builds disclosure tooling for companies and financial institutions in Australia and Singapore — the GHG inventory, the ISSB and AASB disclosure structures, scenario analysis, and PCAF-aligned financed-emissions treatment with data quality scoring. The tools are free to use and require no account to start.

WEB

regenesisimpact.in

CORRESPONDENCE

info@regenesisimpact.in

Corrections are welcome. This briefing describes a standard that is actively being revised; if a requirement or figure has moved, please write and it will be amended.

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